- *Security*—Protecting against potential misuse of generative capabilities for malicious purposes, such as generating fake content, spreading misinformation, or impersonating individuals, is paramount. These trespasses can have a long-lasting and damaging effect on public opinion.
- *Regulatory compliance*—It's imperative to adhere to regulations and guidelines related to data privacy, intellectual property rights, and content authenticity so your tools don't infringe on copyrighted material or create deepfakes. You need to understand and comply with regulations such as the General Data Protection Regulation (GDPR) in Europe and the California Consumer Privacy Act (CCPA) in the United States. These laws dictate how you can collect, store, and process user data, and they are especially relevant when dealing with personally identifiable information (PII). PII is any data that can be used to identify a specific individual, such as names, addresses, phone numbers, and email addresses. You must have a clear strategy for handling PII, including obtaining user consent, securely storing the data, and providing a mechanism for users to request its deletion.

1.6.2 Will developers lose jobs because of AI?

Yes, that is a possibility—but only if we fail to harness this technology to enhance our skills. Imagine you're a skilled furniture maker during the Industrial Revolution, sought after for your craftsmanship. Then, automated machines flood the market, promising to streamline the production of quality furniture at a lower cost. Initially, you might be skeptical of the threat, thinking, "My craft is too intricate to be automated, so I'm safe." But as the machines get better, you may fear that they'll replace you entirely. Notice, however, that two centuries later, there are still carpenters around. Successful furniture makers adapted to technology, learned new skills, and found new opportunities to innovate.

Still, it's undeniable that many jobs were affected by this change and have become obsolete—handloom weavers, blacksmiths, coopers (barrel makers), spinners and carders (textile workers), shoemakers (cobblers), candlemakers, and scribes (copyists). Similarly, in software development, tasks like code generation, documentation, test-case creation, and reviews may become highly automated. Yet, this transformation presents an opportunity for developers to focus on more meaningful and creative aspects of their work. If you aspire to remain relevant and competent in your craft as a developer, my recommendation is to embrace new and innovative technologies like generative AI.

1.6.3 Are generative AI outputs reliable?

As we've seen, generative AI tools are based on natural language, machine learning models that infer their next output based on the set of highest probabilities. This approach introduces two immediate questions:

- How do we trust those probabilities to be consistently accurate?
- How can we address potential language bias in the outputs based on the training data?